Last Time

Finishing Up Distributions

QQ Plots

Introduction to Statistical Inference

The Central Limit Theorem

Sampling Distribution of $\overline{X}$

Confidence Intervals

When σ^2 is Unknown

Today

Sampling Distribution of $\overline{X}$

The Central Limit Theorem

Confidence Intervals

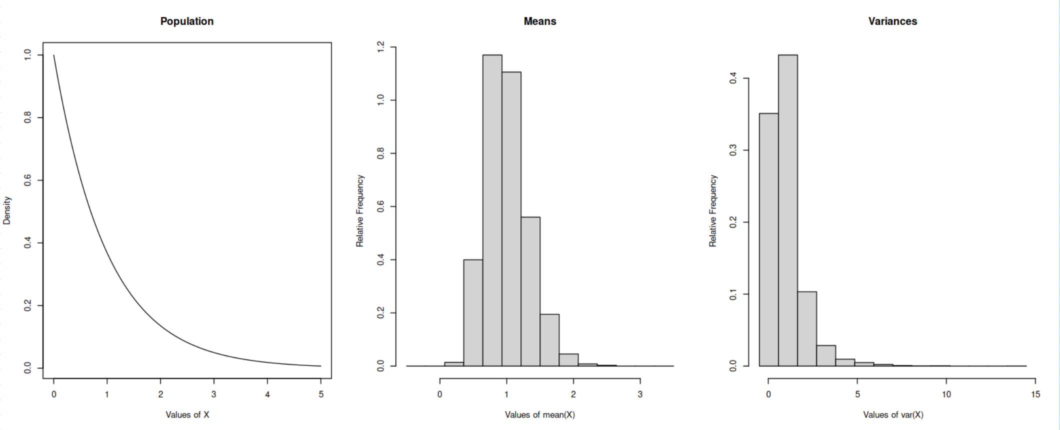
When σ^2 is Unknown

Other Confidence Intervals

Sampling Distributions

```
# Draw samples from an Exp(1) population
plot(seq(0,5,length=200),dexp(seq(0,5,length=200)),
     type ='l',xlab = "Values of X",ylab="Density")

bignum <- 10000
N <- 10
means <- vector()
vars <- vector()

for (i in 1:bignum) {
  X <- rexp(N)
  means[i] <- mean(X)
  vars[i] <- var(X)
}

par(mfrow=c(1,2))
hist(means,breaks=seq(-.5,round(max(means))+.5,length=15),freq=FALSE,
     ylab="Relative Frequency",xlab="Values of mean(X)",main="Means")

hist(vars,breaks=seq(-.5,round(max(vars))+.5,length=15),freq=FALSE,
     ylab="Relative Frequency",xlab="Values of var(X)",
     main="Variances")
```

Sampling Distributions



Sampling Distribution

Definition (Sampling Distribution)

A *sampling distribution* is the distribution of possible values of $S(X)$ under all possible random samples of X of a fixed size N.

Distributions of Statistics

Single samples of size N

Parameter	Test Statistic	Distribution
μ	$\overline{X}$	$\mathcal{N}(\mu, \sigma^2/N)$
μ	$\dfrac{\overline{X} - \mu}{s/\sqrt{N}}$	$t(N-1)$
π	p	$\mathcal{N}(\pi, \pi(1-\pi)/N)$
σ^2	$\dfrac{(N-1)s^2}{\sigma^2}$	$\chi^2(N-1)$
ρ	$\frac{1}{2}\ln[(1+r)/(1-r)]$	$\mathcal{N}(\tanh^{-1}(\rho), 1/\sqrt{N-3})$

The Central Limit Theorem

Consider any variable X and a sample of size N of that variable.
As N gets large (goes to infinity), the shape of the distribution of the sample mean $\overline{X}$ will become normal.

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Formally: For an iid sample $X = \{X_1, X_2, \ldots, X_N\}$ of size N drawn from a population with mean μ and standard deviation σ, define the random variable $S_N = \sum_{i=1}^{N} X_i$.

$$\lim_{N \to \infty} Pr\left(\frac{S_N - N\mu}{\sigma\sqrt{N}} \leq x\right) = P(Z \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-z^2/2} dz.$$

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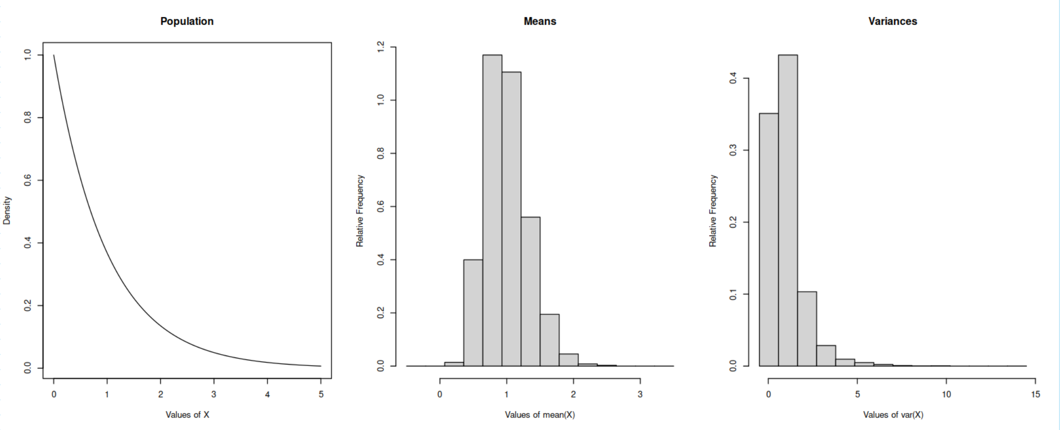
Regardless of the shape of the distribution of X!

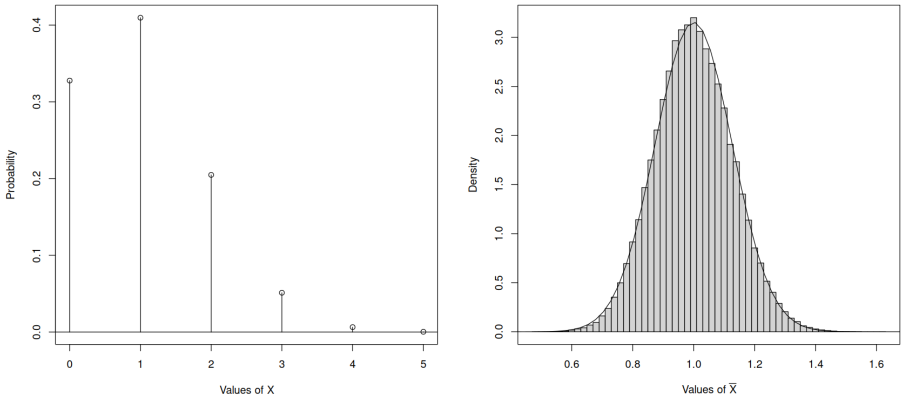
Exponential Population



Central Limit Theorem

```
# Construct again the sampling distribution of the mean, samples drawn from a
# Binomial(5,.2) distribution
bignum <- 100000 # infinity

# Binomial parameters and sample size
N <- 5
p <- .2
sample.size <- 50

# Use replicate() to obtain the means
x <- replicate(bignum,mean(rbinom(sample.size,N,p)))

# Plotting commands
par(mfrow=c(1,2))
# Binomial population
plot(seq(0,5),dbinom(0:5,N,p),type='p',
     xlab="Values of X",ylab="Probability")
segments(x0=0:5,y0=0,x1=0:5,y1=dbinom(0:5,N,p))
abline(h=0)

# Sampling distribution of mean
hist(x,probability=TRUE,breaks=seq(.47,1.63,by=.02),main="",
     xlab=expression("Values of "*bar(X)))
mean.values <- seq(0,N,length=200)
lines(mean.values,
      dnorm(mean.values,N*p,sqrt(N*p*(1-p)/sample.size)))
box()
```

Central Limit Theorem



The Sampling Distribution of the Mean

By the CLT, we know $\overline{X} \sim \mathcal{N}(\mu_{\overline{X}}, \sigma^2_{\overline{X}})$

What are $\mu_{\overline{X}}$ and $\sigma^2_{\overline{X}}$?

$\mu_{\overline{X}}$

$$E(\overline{X}) = E\left(\frac{1}{N}\sum_{i=1}^{N} X_i\right)$$

$\mu_{\overline{X}}$

$$
\begin{aligned}
E(\overline{X}) &= E\left(\frac{1}{N}\sum_{i=1}^{N} X_i\right) \\
&= \frac{1}{N} E\left(\sum_{i=1}^{N} X_i\right) \\
&= \frac{1}{N}\left(\sum_{i=1}^{N} E(X_i)\right) \\
&= \frac{1}{N}\left(\sum_{i=1}^{N} \mu\right) \\
&= \frac{1}{N} N\mu = \mu.
\end{aligned}
$$

$\sigma^2_{\overline{X}}$

$$Var(\overline{X}) = Var\left(\frac{1}{N}\sum_{i=1}^{N} X_i\right)$$

$\sigma^2_{\overline{X}}$

$$
\begin{aligned}
Var(\overline{X}) &= Var\left(\frac{1}{N}\sum_{i=1}^{N} X_i\right) \\
&= \frac{1}{N^2} Var\left(\sum_{i=1}^{N} X_i\right) \\
&= \frac{1}{N^2}\left(\sum_{i=1}^{N} Var(X_i)\right) \\
&= \frac{1}{N^2}\left(\sum_{i=1}^{N} \sigma^2\right) \\
&= \frac{1}{N^2} N\sigma^2 = \frac{\sigma^2}{N}.
\end{aligned}
$$

Standard Deviation vs. Standard Error

- The standard deviation σ refers to the population X.
- The standard error $\sigma_{\overline{X}} = \sigma/\sqrt{N}$ is the standard deviation of $\overline{X}$.
- The standard error is still a standard deviation, but calling it standard error focuses our attention on how well $\overline{X}$ estimates the population mean μ.

Estimating μ

- We have $\overline{X}$; we want μ.
- We know that $\overline{X} \sim \mathcal{N}(\mu, \sigma^2/N)$
  1. $\overline{X}$ is a point estimate of μ: $\overline{X} = \hat{\mu}$.
  2. Using what we know about the distribution of $\overline{X}$ we can construct an interval that *covers* μ with some selected probability.

Point Estimate

What's the best single, numerical guess for μ?

- $\overline{X} = \hat{\mu}$
- Consistent: Law of Large Numbers: $\lim_{N \to \infty} \overline{X} = \mu$
- Unbiased: $E(\overline{X}) = \mu$
- Other good properties

Interval Estimate

- Assume we know what σ^2 is.
- An *interval estimate* is a range of possible values for μ.
- Based on the sampling distribution for $\overline{X}$, we can determine the width of an interval that will include μ some specified percentage of the time.

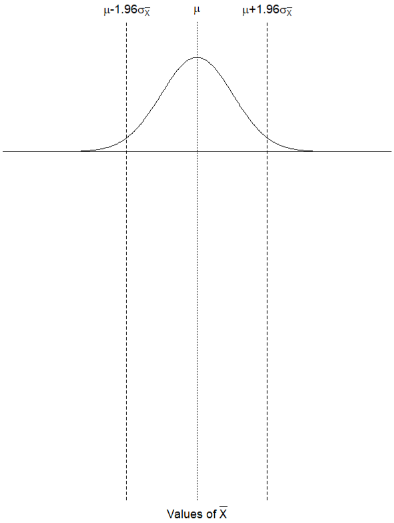
Back to the Normal Distribution

Consider a standard normal Z random variable.

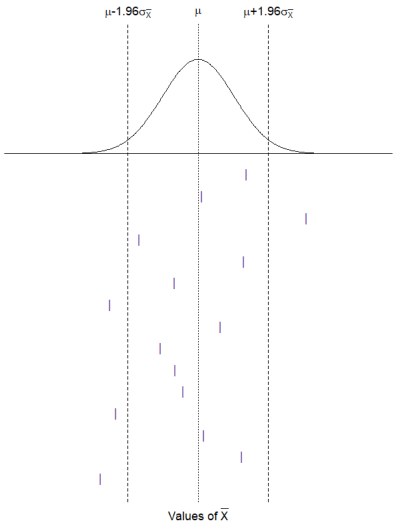
Back to the Normal Distribution

Consider a standard normal Z random variable.
Place upper and lower bounds symmetrically around $\mu_Z = 0$.
Suppose we want the probability of sampling a z-score between those boundaries to be 95%. What are the values of the boundaries?

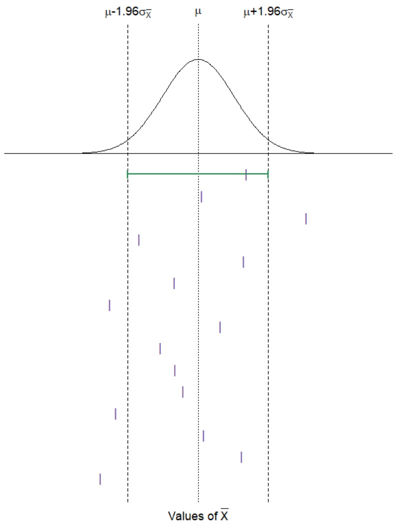
Understanding Confidence Intervals



Understanding Confidence Intervals



Understanding Confidence Intervals



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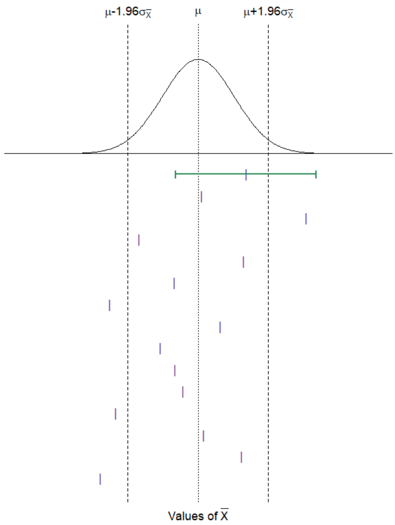
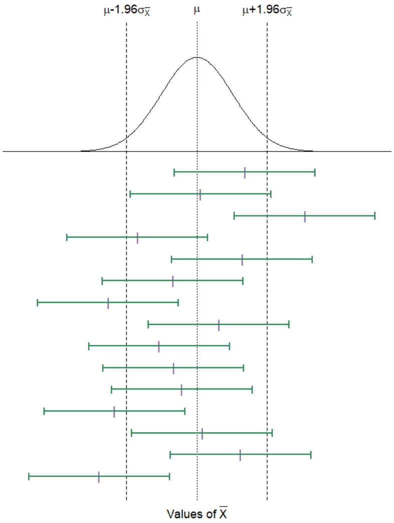


Understanding Confidence Intervals



What Is Random?

The interval $\overline{X} \pm 1.96\sigma_{\overline{X}}$ will cover μ for 95% of the values of $\overline{X}$ that we observe. What is random?

- $\overline{X}$
- σ
- μ
- Interval $[L, U]$
- Interval width $U - L$

Another Way To Think About It

How do we go from an interval around $Z_{\overline{X}}$ to an interval around $\overline{X}$?

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$$
\begin{aligned}
& Pr(-1.96 < Z_{\overline{X}} < 1.96) \\
= \quad & .95
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$$
\begin{aligned}
&Pr(-1.96 < Z_{\overline{X}} < 1.96) \\
&= \quad .95 \\
&= \quad Pr\left(-1.96 < \frac{\overline{X} - \mu}{\sigma_{\overline{X}}} < 1.96\right) \\
&= \quad Pr(-1.96\sigma_{\overline{X}} < \overline{X} - \mu < 1.96\sigma_{\overline{X}}) \\
&= \quad Pr(-\overline{X} - 1.96\sigma_{\overline{X}} < -\mu < -\overline{X} + 1.96\sigma_{\overline{X}}) \\
&= \quad Pr(\overline{X} + 1.96\sigma_{\overline{X}} > \mu > \overline{X} - 1.96\sigma_{\overline{X}}) \\
&= \quad Pr(\overline{X} - 1.96\sigma_{\overline{X}} < \mu < \overline{X} + 1.96\sigma_{\overline{X}})
\end{aligned}
$$

But What Is Random Again?

$$Pr(\overline{X} - 1.96\sigma_{\overline{X}} < \mu < \overline{X} + 1.96\sigma_{\overline{X}}) = 0.95$$

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$$Pr(\overline{X} - 1.96\sigma_{\overline{X}} < \mu < \overline{X} + 1.96\sigma_{\overline{X}}) = 0.95$$

Suppose $\overline{X} = 97$ and $\sigma_{\overline{X}} = 4$. Then

$$\begin{aligned} \overline{X} - 1.96\sigma_{\overline{X}} &= 89.16 \\ \overline{X} + 1.96\sigma_{\overline{X}} &= 104.84 \end{aligned}$$

Do we write $Pr(89.16 < \mu < 104.84) = 0.95$?

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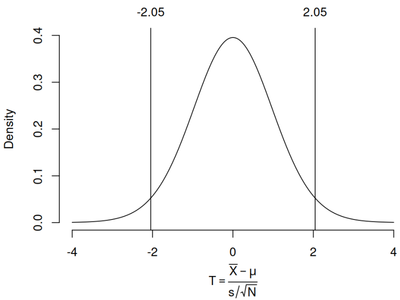
Do we write $Pr(89.16 < \mu < 104.84) = 0.95$?

APA Publication Manual:
Write "$\overline{X} = 97,\ 95\%\ CI\ [89.16, 104.84]$"

σ^2 is Unknown

$$T = \frac{\overline{X} - \mu}{s/\sqrt{N}} \sim t(N-1)$$



σ^2 is Unknown

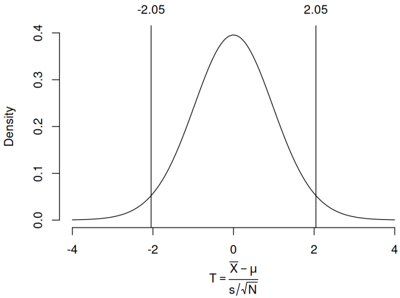
$$T = \frac{\overline{X} - \mu}{s/\sqrt{N}} \sim t(N-1)$$

Z-interval:

$$[\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{N}}, \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{N}}]$$

T-interval:

$$[\overline{X} - t_{\alpha/2}\frac{s}{\sqrt{N}}, \overline{X} + t_{\alpha/2}\frac{s}{\sqrt{N}}]$$



Normally distributed X

```
# Confidence intervals
# Draw 10000 samples of size 30 from an exponential distribution
# with mean 5 and compute the sample means
bignum <- 10000 # Number of samples
N <- 30         # Sample size
lambda <- 5     # Population mean
conf <- .95     # Confidence level

# Sample from an Exp(lambda) distribution
x <- matrix(rexp(30*10000,1/lambda),ncol=30)

# Compute the means and standard errors for each sample
means <- rowMeans(x)
ses <- apply(x,1,sd)/sqrt(N) # Sample sd divided by sqrt(N)

# Variance known: sigma^2 = 25
z.limits <- qnorm(c((1-conf)/2,1-(1-conf)/2))
z.lower <- means + z.limits[1]*lambda/sqrt(N)
z.upper <- means + z.limits[2]*lambda/sqrt(N)

# Variance unknown: use s^2
t.limits <- qt(c((1-conf)/2,1-(1-conf)/2),N-1)
t.lower <- means + t.limits[1]*ses
t.upper <- means + t.limits[2]*ses

# Compute the probabilities that the z and t intervals
# contain lambda
conf.z <- mean(lambda >= z.lower & lambda <= z.upper)
conf.t <- mean(lambda >= t.lower & lambda <= t.upper)

rbind(c("Confidence","Sigma known","Sigma unknown"),
      c(0.95,round(conf.z,2),round(conf.t,2)))
```

Confidence Interval Around the Sample Variance

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\frac{(N-1)s^2}{\sigma^2} \sim \chi^2(N-1)$$

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$$
\begin{aligned}
& Pr(\chi^2_{.025}(N-1) \leq \frac{(N-1)s^2}{\sigma^2} \leq \chi^2_{.975}(N-1)) \\
= \quad & 0.95 \\
= \quad & Pr(\frac{1}{\chi^2_{.975}(N-1)} \leq \frac{\sigma^2}{(N-1)s^2} \leq \frac{1}{\chi^2_{.025}(N-1)}) \\
= \quad & Pr(\frac{(N-1)s^2}{\chi^2_{.975}(N-1)} \leq \sigma^2 \leq \frac{(N-1)s^2}{\chi^2_{.025}(N-1)}).
\end{aligned}
$$

Confidence Interval Around the Sample Variance